Software Requirements Specification (SRS)

IoT Based Sleep Monitoring for Eco-Friendly Pain Relief Pillow

Version 1.0

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Project Guide

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Revision History

Version	Date	Author(s)	Description of Changes
0.1	23 July 2026	Project Team	Initial concept and project proposal
0.5	24 July 2026	Project Team	Hardware architecture finalized
0.8	25 July 2026	Project Team	Functional requirements drafted
1.0	30 July 2026	Project Team	Initial SRS released for review

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the **Low-Cost IoT Sleep Monitoring Pillow Prototype**. The document serves as the primary reference for system design, hardware integration, software development, testing, and validation of the proposed smart pillow monitoring system.

The objective of the prototype is to evaluate the effectiveness of an organically developed pillow by collecting sleep-related parameters using affordable Internet of Things (IoT) technologies. The acquired data are intended to support comparative analysis of sleep comfort before and after using the proposed pillow.

The system is designed exclusively for academic research and product evaluation purposes. It is not intended to diagnose sleep disorders or replace clinical sleep-monitoring equipment.

1.2 Introduction

Sleep quality plays an important role in physical health, mental well-being, and daily performance. Improper pillow support can contribute to neck pain, discomfort, restlessness, and poor sleep quality, while conventional sleep-monitoring methods are often costly and unsuitable for continuous use in natural home environments. Recent studies on smart pillows have shown that low-cost IoT-based systems can provide a practical and non-intrusive approach for monitoring sleep-related behavior such as head movement, posture, and environmental conditions.

The proposed project aims to develop and evaluate an organic material pillow prepared using natural ingredients such as rice grass, Jatamansi for relaxation, and processed cow-dung-derived organic material. In addition to improving comfort and relaxation, the pillow also supports sustainability by promoting the use of biodegradable, eco-friendly, and naturally sourced materials as an alternative to conventional synthetic pillow fillings. Organic and eco-friendly pillows are increasingly recognized for their potential to reduce chemical exposure and support environmentally responsible product design.

To assess the effectiveness of the proposed pillow, this project integrates a low-cost IoT-based sleep monitoring system capable of collecting before-and-after sleep data. The monitoring setup acquires data related to heart rate, breathing patterns, pillow pressure distribution, head movement, and pillow surface temperature. These measurements enable comparative evaluation of sleep quality, comfort, neck pain, and restlessness while maintaining a completely non-clinical and non-invasive monitoring approach.

By combining sustainable material innovation with smart sensing technologies, the proposed system contributes to both environmentally responsible product development and affordable IoT-based sleep research.

1.3 Scope

The proposed system is a non-clinical IoT-based sleep monitoring prototype intended for comparative analysis of sleep behavior in natural home environments or controlled academic research settings. The system continuously acquires physiological and environmental data during sleep using low-cost sensors integrated with an ESP32-based data acquisition unit.

The prototype utilizes four sensing mechanisms:

- A wearable non-invasive heart rate sensor for monitoring cardiovascular activity.
- A microphone-based breathing sensor that measures breathing intensity through sound level analysis without recording or storing voice data.
- A Velostat-based pressure sensing matrix positioned beneath the pillow to monitor pressure distribution, head position, and movement frequency.
- A temperature sensor to observe pillow surface temperature and evaluate the thermal behavior of the proposed organic pillow.

The collected data are transmitted to a cloud database for storage, visualization, and comparative analysis. The system enables researchers to compare sleep-related parameters before and after using the proposed organic pillow, thereby assisting in evaluating improvements in comfort, reduction in movement, thermal characteristics, and perceived sleep quality.

The prototype is intended solely for research, educational, and product evaluation purposes. It does not perform medical diagnosis, sleep-stage classification, or treatment recommendation.

1.4 Objectives

The primary objectives of the proposed system are:

1. To design and develop a low-cost IoT-based sleep monitoring pillow prototype using an ESP32 microcontroller and affordable sensing technologies.
2. To monitor heart rate during sleep using a wearable, non-invasive wrist-based heart rate sensor.
3. To capture breathing patterns using a microphone by analyzing breathing sound intensity without recording or storing voice data, thereby ensuring user privacy.
4. To develop a Velostat-based pressure sensing matrix capable of detecting head position, pressure distribution, and movement frequency throughout the sleep session.
5. To monitor pillow surface temperature and evaluate the thermal characteristics of the proposed organic pillow in comparison with conventional pillows.
6. To collect and compare sleep-related data before and after using the proposed organic pillow, including movement frequency, pressure variation, breathing patterns, heart rate trends, temperature variation, and sleep duration.
7. To assess improvements in sleep comfort, reduction in neck pain, and decrease in sleep disturbances through sensor observations combined with user feedback.

8. To evaluate whether the proposed organic pillow provides measurable improvements in sleep quality, comfort, and thermal performance when compared with a conventional pillow.

1.5 Definitions, Acronyms and Abbreviations

Term	Definition
IoT	Internet of Things
ESP32	Low-power Wi-Fi and Bluetooth-enabled microcontroller used for data acquisition
HR	Heart Rate
BPM	Beats Per Minute
dB	Decibel
Velostat	Pressure-sensitive conductive material used for force sensing
Wi-Fi	Wireless communication protocol used for cloud connectivity
BLE	Bluetooth Low Energy
CSV	Comma Separated Values
JSON	JavaScript Object Notation
RTC	Real-Time Clock
UI	User Interface

1.6 References

1. IEEE Std 29148-2018 – *Systems and Software Engineering – Life Cycle Processes – Requirements Engineering.*
2. IEEE Std 830-1998 – *Recommended Practice for Software Requirements Specifications.*
3. Espressif Systems. *ESP32 Series Technical Reference Manual.*
4. Relevant research articles on smart pillow systems, IoT-based sleep monitoring, Velostat pressure sensing, and sustainable ergonomic pillow design.

1.7 Document Organization

This document is organized into seven major sections. Following the introduction, the overall system description explains the product perspective, operating environment, user classes, and design constraints. Subsequent sections describe the system architecture, hardware and software interfaces, functional requirements, non-functional requirements, and verification methodology. The appendices provide supplementary information, including sensor specifications, communication protocols, block diagrams, and testing procedures.

2. Overall Description

2.1 Product Perspective

The **Low-Cost IoT Sleep Monitoring Pillow Prototype** is a non-clinical smart monitoring system designed to evaluate the effectiveness of an organic ergonomic pillow through continuous observation of sleep-related parameters. The prototype combines low-cost sensing technologies, wireless communication, cloud-based storage, and data visualization to provide researchers with quantitative information regarding sleep comfort and pillow performance.

Unlike conventional polysomnography (PSG) systems, which require multiple clinical sensors and specialized laboratory environments, the proposed system adopts a simple, affordable, and non-invasive approach suitable for home environments and academic research. The system does not attempt to diagnose sleep disorders or classify sleep stages. Instead, it records physiological and physical parameters that are associated with sleep comfort, head movement, breathing behavior, and thermal characteristics.

The prototype consists of four sensing modules integrated with an ESP32-based data acquisition unit:

- A wearable heart rate sensor to monitor cardiovascular activity during sleep.
- A microphone-based breathing sensor that detects breathing intensity through sound level analysis while ensuring that no voice recordings are stored.
- A Velostat-based pressure sensing matrix placed beneath the pillow to measure pressure distribution, head position, and movement frequency.
- A temperature sensor to monitor pillow surface temperature and evaluate the thermal behavior of the organic pillow.

The ESP32 acquires data from the sensors, performs preliminary processing, timestamps the measurements, and transmits them to a cloud database through Wi-Fi. A dashboard application retrieves the stored data to generate visualizations and comparative reports for before-and-after pillow evaluation.

2.2 Product Functions

The primary functions of the proposed system are summarized below.

Heart Rate Monitoring

The system continuously receives heart rate measurements from a wearable wrist-based heart rate sensor. The acquired data are stored with timestamps for later visualization and comparison between different sleep sessions.

Breathing Pattern Monitoring

A microphone positioned near the pillow captures breathing sound intensity by measuring decibel levels. The microphone does not record speech or voice data. Instead, only processed sound intensity values are stored, enabling breathing pattern visualization while maintaining user privacy.

Pressure Distribution Monitoring

A Velostat-based pressure sensing matrix installed beneath the pillow detects the pressure exerted by the user's head. The pressure values are used to determine head position, pressure distribution, and variations occurring throughout the sleep session.

Head Movement Detection

Changes in pressure distribution over time are analyzed to identify head movements and movement frequency. These observations provide an indication of sleep restlessness and user comfort.

Temperature Monitoring

A temperature sensor continuously monitors the pillow surface temperature. The recorded thermal profile is used to evaluate the heat retention and cooling characteristics of the proposed organic pillow compared with conventional pillows.

Data Acquisition and Storage

The ESP32 periodically collects sensor readings, timestamps the measurements, and transmits them to a cloud database using Wi-Fi. The stored information can later be exported for detailed analysis.

Data Visualization

A web-based or mobile dashboard displays graphs of heart rate, breathing intensity, pressure distribution, movement events, and temperature variation throughout the sleep session.

Comparative Sleep Analysis

The collected data are compared between baseline sleep sessions using a conventional pillow and subsequent sessions using the proposed organic pillow. This comparison assists researchers in evaluating improvements in comfort, reduction in movement, and thermal performance.

2.3 User Classes and Characteristics

The proposed system is intended for the following categories of users.

2.3.1 Test Subject

The test subject is the individual sleeping on the pillow during data collection. The user interacts minimally with the system by wearing the heart rate sensor and sleeping naturally. No manual operation is required during the monitoring period.

2.3.2 Researcher

Researchers configure the system, initiate data collection, monitor sensor status, retrieve recorded datasets, and perform comparative analysis. They also interpret graphical reports generated by the dashboard.

2.3.3 Project Developers

The development team is responsible for integrating hardware, maintaining firmware, improving algorithms, calibrating sensors, and troubleshooting communication or software-related issues.

2.3.4 Faculty Supervisor

The faculty supervisor evaluates project progress, reviews experimental data, verifies system performance, and validates the effectiveness of the proposed organic pillow.

2.4 Operating Environment

The proposed system operates within the following environment.

Component	Description
Microcontroller	ESP32 Development Board
Heart Rate Sensor	Wearable wrist-based sensor (Bluetooth-enabled or equivalent)
Breathing Sensor	Electret or MEMS microphone with sound intensity processing
Pressure Sensor	Velostat-based pressure sensing matrix
Temperature Sensor	Contact temperature sensor (e.g., DS18B20, LM35, or equivalent)
Communication	Wi-Fi (IEEE 802.11 b/g/n)
Cloud Platform	Firebase, ThingSpeak, AWS IoT, or equivalent
Database	Cloud-hosted real-time database
Dashboard	Web or mobile application

Development Environment	Arduino IDE or ESP-IDF
Programming Language	C/C++, JavaScript/Python (for dashboard and analytics)

2.5 Design and Implementation Constraints

The development of the proposed prototype is subject to the following constraints.

C-1 Low-Cost Design

The system shall utilize affordable and commercially available hardware components to ensure that the overall prototype remains economically feasible for academic research.

C-2 Non-Invasive Operation

The monitoring process shall not interfere with the user's normal sleeping behavior. All sensors shall operate without causing discomfort or restricting movement.

C-3 Privacy Protection

The breathing sensor shall analyze only sound intensity levels. The system shall not record, store, or transmit speech or voice recordings.

C-4 Wireless Communication

Sensor data shall be transmitted wirelessly through the ESP32 using Wi-Fi. Temporary communication failures shall not result in permanent loss of recorded data whenever local buffering is available.

C-5 Research Prototype

The system is intended solely for research and comparative evaluation purposes and shall not provide medical diagnosis or treatment recommendations.

C-6 Eco-Friendly Integration

The embedded sensing system shall be designed so that it does not significantly alter the ergonomic properties or sustainability objectives of the proposed organic pillow.

2.6 Assumptions and Dependencies

The following assumptions apply during system operation.

- The wearable heart rate sensor remains properly worn throughout the sleep session.
- The microphone is positioned to primarily capture breathing sounds while minimizing ambient environmental noise.
- The Velostat pressure sensing matrix remains correctly aligned beneath the pillow during experimentation.
- The temperature sensor maintains consistent contact with the pillow surface for accurate measurements.
- Wi-Fi connectivity is available for uploading recorded sensor data to the cloud database.
- The ESP32 remains continuously powered throughout the monitoring period.
- Users perform experiments under similar environmental conditions during before-and-after comparisons to ensure consistency.

3. System Architecture

3.1 Overall System Architecture

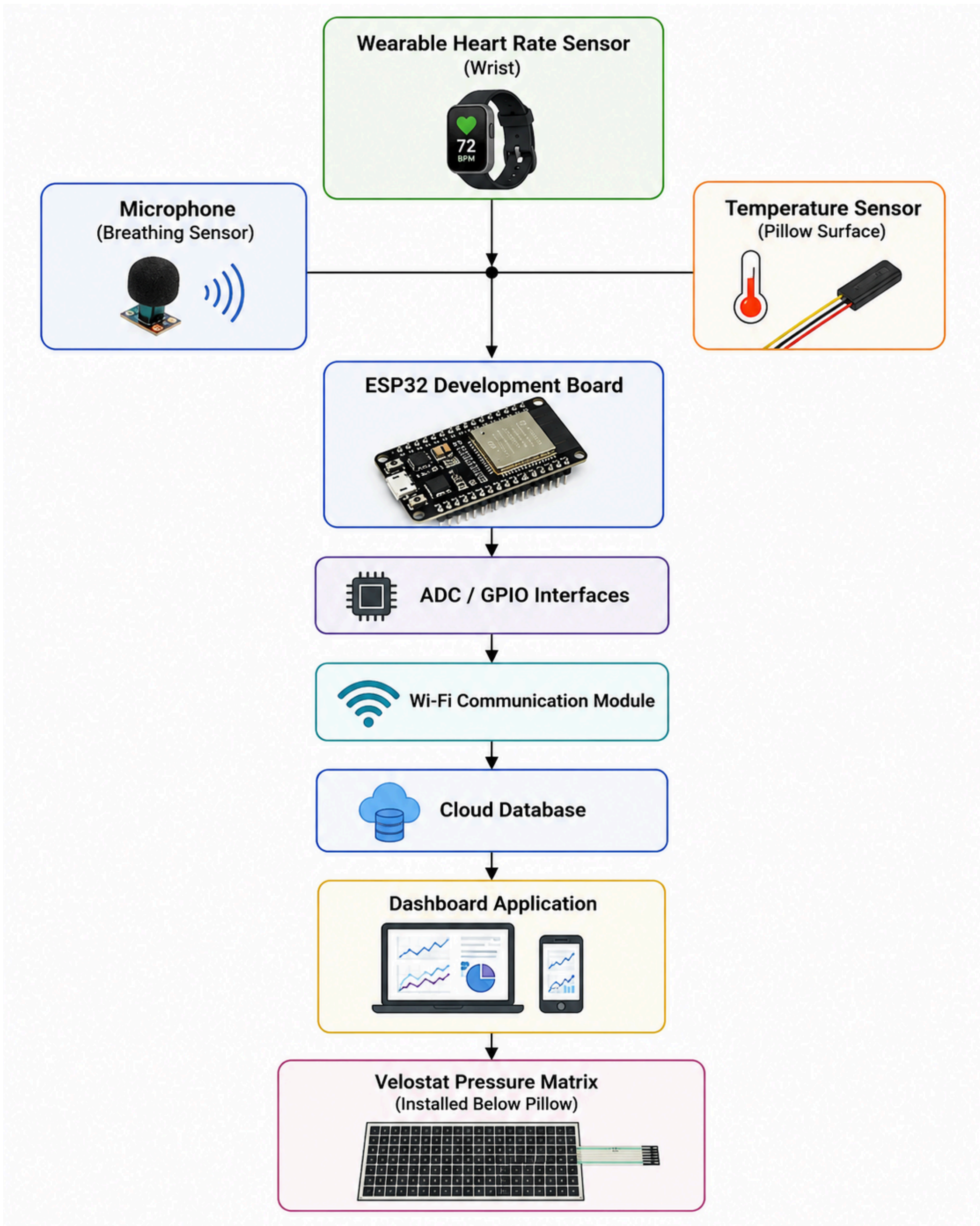
The proposed prototype follows a layered IoT architecture consisting of four primary layers:

1. **Sensing Layer**
2. **Edge Processing Layer**
3. **Cloud Layer**
4. **Application Layer**

Sensor data are acquired in real time, processed by the ESP32, transmitted to the cloud, and finally visualized through a dashboard for comparative analysis.

3.2 Hardware Architecture

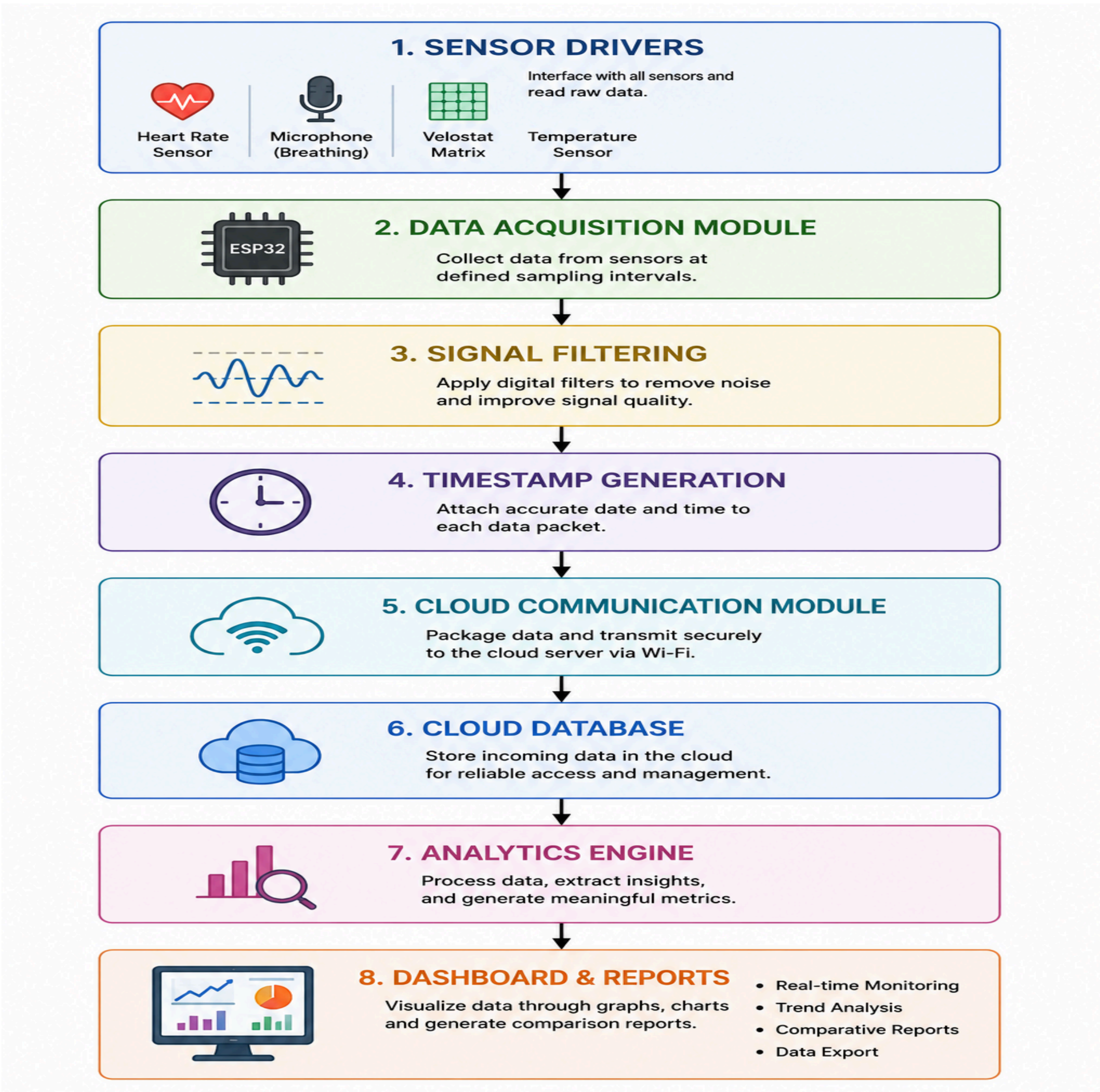
The hardware architecture consists of four sensing modules connected to an ESP32 microcontroller.



Wearable Heart Rate Sensor
(Wrist)
72
BPM
Microphone
(Breathing Sensor)
Temperature Sensor
(Pillow Surface)
ESP32 Development Board
ADC / GPIO Interfaces
Wi-Fi Communication Module
Cloud Database
Dashboard Application
Velostat Pressure Matrix
(Installed Below Pillow)

3.3 Software Architecture

The software is divided into multiple functional modules.



3.4 Data Flow

The operational sequence of the proposed system is as follows:

1. The heart rate sensor, microphone, Velostat pressure matrix, and temperature sensor collect sleep-related data.
2. The ESP32 receives sensor readings at predefined sampling intervals.
3. The acquired measurements are filtered and timestamped.
4. The processed data are transmitted to the cloud database via Wi-Fi.
5. The dashboard retrieves stored data and generates graphs and comparative reports.
6. Researchers analyze the differences between baseline and organic pillow sleep sessions to evaluate improvements in comfort and sleep behavior.

3.5 Sensor Interfaces

Sensor	Parameter Measured	Interface	Purpose
Wearable Heart Rate Sensor	Heart Rate (BPM)	Bluetooth Low Energy (BLE) or Serial	Monitor cardiovascular activity during sleep
Microphone	Breathing sound intensity (dB)	Analog/ADC	Analyze breathing patterns without storing voice data
Velostat Pressure Matrix	Pressure distribution and head movement	Analog Multiplexed Inputs	Detect head position, pressure variation, and movement frequency
Temperature Sensor	Pillow surface temperature (°C)	Digital (One-Wire/I^2C) or Analog	Evaluate thermal behavior and cooling characteristics of the pillow

4. External Interface Requirements

4.1 Hardware Interfaces

The proposed Low-Cost IoT Sleep Monitoring Pillow Prototype interfaces with four sensing devices and an ESP32-based data acquisition unit. Each sensor measures a specific physiological or environmental parameter related to sleep comfort and pillow performance. The ESP32 serves as the central controller responsible for sensor communication, data processing, and wireless transmission.

HIR-1 Wearable Heart Rate Sensor Interface

The system shall interface with a wearable, non-invasive wrist-based heart rate sensor capable of continuously measuring heart rate during sleep. The heart rate sensor shall communicate with the ESP32 using Bluetooth Low Energy (BLE) or an equivalent wireless communication protocol.

Input Parameter: Heart Rate (BPM)

Communication Interface: Bluetooth Low Energy (BLE)

Purpose: Continuous monitoring of heart rate trends during sleep.

HIR-2 Breathing Sensor Interface

The breathing sensor consists of a sensitive microphone positioned near the pillow. The microphone measures breathing sound intensity by detecting variations in sound pressure level. The system processes only the decibel values and breathing waveform characteristics without recording or storing voice data.

Input Parameter: Breathing Sound Intensity (dB)

Communication Interface: Analog Input (ADC)

Purpose: Monitoring breathing patterns while maintaining user privacy.

HIR-3 Velostat Pressure Sensor Interface

The pressure sensing module consists of a Velostat-based pressure matrix embedded beneath the pillow. Variations in resistance caused by applied pressure are measured through the ESP32 Analog-to-Digital Converter (ADC). The acquired pressure map is used to estimate head position and detect movement frequency.

Input Parameters:

- Pressure Distribution
- Head Position
- Pressure Variation
- Movement Frequency

Communication Interface: Analog Multiplexed Inputs (ADC)

HIR-4 Temperature Sensor Interface

A temperature sensor is placed near the pillow surface to continuously monitor surface temperature throughout the sleep session. The recorded temperature profile is used to evaluate heat retention and cooling characteristics of the proposed organic pillow.

Input Parameter: Pillow Surface Temperature (°C)

Communication Interface: One-Wire / I²C / Analog (depending on selected sensor)

4.2 Software Interfaces

The software architecture consists of firmware executing on the ESP32, cloud services for data storage, and a dashboard application for visualization and analysis.

SIR-1 Firmware

The firmware shall perform:

- Sensor initialization
- Periodic sensor data acquisition
- Signal preprocessing
- Timestamp generation
- Data packet formation
- Wireless communication

SIR-2 Cloud Database

The cloud database shall store all sensor observations received from the ESP32.

Each uploaded record shall include:

- Timestamp
- Heart Rate
- Breathing Intensity
- Pressure Sensor Values
- Temperature
- Device Identifier

SIR-3 Dashboard

The dashboard shall retrieve stored data and display:

- Heart Rate Graph
- Breathing Pattern Graph
- Pressure Distribution Graph
- Head Movement Timeline
- Temperature Variation Graph
- Comparative Before-and-After Reports

4.3 Communication Interfaces

Communication between system components shall follow the architecture shown in Section 3.

CIR-1 Sensor Communication

Sensor	Communication
Heart Rate Sensor	Bluetooth Low Energy (BLE)
Microphone	Analog ADC
Velostat Matrix	Analog ADC
Temperature Sensor	One-Wire / I²C / Analog

CIR-2 ESP32 to Cloud Communication

The ESP32 shall transmit sensor readings to the cloud database through Wi-Fi using HTTP or MQTT communication protocols.

Each transmitted packet shall contain:

- Device ID
- Timestamp
- Heart Rate
- Breathing Intensity
- Pressure Values
- Temperature

CIR-3 Data Synchronization

Each sensor reading shall be timestamped before transmission to maintain synchronization among multiple sensor streams.

4.4 Database Interfaces

The cloud database shall organize sensor observations according to sleep sessions.

Each session shall include:

Field	Description
Session ID	Unique Sleep Session Identifier
User ID	Participant Identifier

Timestamp	Date and Time
Heart Rate	BPM
Breathing Intensity	dB
Pressure Data	Velostat Matrix Values
Temperature	Pillow Temperature
Remarks	Optional Notes

The database shall support:

- Real-time data insertion
- Historical data retrieval
- CSV export
- Comparative session analysis

4.5 User Interface

The dashboard shall provide an intuitive graphical interface for researchers.

The interface shall display:

Home Screen

- Device Connection Status
- Current Sensor Status
- Active Sleep Session

Live Monitoring Screen

- Heart Rate
- Breathing Intensity
- Pressure Heat Map
- Temperature
- Wi-Fi Status

Historical Analysis

Users shall be able to:

- Select previous sleep sessions
- Compare baseline and experimental sessions

- View graphical trends
- Export reports

Dashboard Outputs

The dashboard shall generate:

- Line graphs
- Pressure heat maps
- Temperature curves
- Heart rate trends
- Movement frequency charts
- Comparative analysis reports

5. Functional Requirements

The functional requirements define the expected behavior of each subsystem in the proposed sleep monitoring prototype. Each requirement is assigned a unique identifier to facilitate verification and validation.

5.1 Heart Rate Monitoring

FR-HR-001 Sensor Initialization

The system shall establish communication with the wearable heart rate sensor before the start of each monitoring session.

FR-HR-002 Continuous Monitoring

The system shall continuously receive heart rate measurements from the wearable sensor throughout the sleep session.

FR-HR-003 Data Logging

Each heart rate reading shall be stored with an accurate timestamp in the cloud database.

FR-HR-004 Trend Visualization

The dashboard shall display heart rate variations over the entire monitoring session using graphical plots.

5.2 Breathing Pattern Monitoring

FR-BR-001 Microphone Initialization

The system shall initialize the breathing sensor before data acquisition begins.

FR-BR-002 Breathing Detection

The microphone shall measure breathing sound intensity using sound pressure levels without storing voice recordings.

FR-BR-003 Privacy Protection

The system shall process only breathing intensity values and shall not record speech or identifiable audio.

FR-BR-004 Pattern Visualization

The dashboard shall display breathing intensity as a time-varying graph.

5.3 Velostat Pressure Monitoring

FR-PS-001 Pressure Acquisition

The system shall continuously measure pressure values from the Velostat sensing matrix.

FR-PS-002 Pressure Distribution

The software shall generate a pressure distribution map representing the user's head position.

FR-PS-003 Head Movement Detection

Changes in pressure distribution over time shall be analyzed to detect head movement events.

FR-PS-004 Movement Frequency

The system shall calculate the total number of head movement events occurring during a sleep session.

FR-PS-005 Pressure Logging

All pressure measurements shall be stored in the cloud database together with timestamps.

5.4 Temperature Monitoring

FR-TMP-001 Continuous Monitoring

The system shall continuously monitor the pillow surface temperature.

FR-TMP-002 Temperature Logging

Temperature readings shall be recorded together with timestamps.

FR-TMP-003 Temperature Trend

The dashboard shall display temperature variation throughout the sleep session.

5.5 Data Acquisition

FR-DA-001 Sensor Synchronization

The ESP32 shall periodically acquire data from all connected sensors.

FR-DA-002 Timestamp Generation

Each acquired sensor reading shall be assigned a timestamp before storage.

FR-DA-003 Data Validation

Invalid or missing sensor readings shall be identified and logged without interrupting system operation.

FR-DA-004 Data Packaging

Sensor readings shall be combined into a single structured data packet before transmission.

Excellent. Below are the final sections of the SRS, rewritten to match your updated project based on the **Wearable Heart Rate Sensor, Microphone, Velostat Pressure Matrix, Temperature Sensor, and ESP32**.

6. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational characteristics of the proposed **Low-Cost IoT Sleep Monitoring Pillow Prototype**. These requirements ensure that the system performs reliably, securely, efficiently, and remains easy to maintain and expand.

6.1 Performance Requirements

NFR-PER-001 Data Acquisition Performance

The system shall acquire data from all connected sensors at predefined sampling intervals without significant delays.

Acceptance Criteria:

- Continuous sensor data acquisition throughout the monitoring session.
- No interruption during normal operation.

NFR-PER-002 Communication Performance

The ESP32 shall transmit sensor data to the cloud database with minimal latency whenever Wi-Fi connectivity is available.

Acceptance Criteria:

- Data transmission delay shall normally remain below **5 seconds**.

NFR-PER-003 Dashboard Performance

The dashboard shall retrieve and display stored sensor data within an acceptable response time.

Acceptance Criteria:

- Historical graphs shall load within **3 seconds** under normal network conditions.

NFR-PER-004 Storage Performance

The database shall support continuous storage of sleep session data without loss of records.

6.2 Reliability Requirements

NFR-REL-001 Continuous Operation

The system shall operate continuously throughout an entire sleep session (typically 6–10 hours) without requiring manual intervention.

NFR-REL-002 Data Integrity

Each sensor reading shall be timestamped before storage to maintain chronological consistency.

NFR-REL-003 Sensor Failure Handling

If one sensor temporarily fails, the remaining sensors shall continue acquiring and transmitting data.

The system shall generate an appropriate sensor status message.

NFR-REL-004 Communication Recovery

Following temporary Wi-Fi interruption, the ESP32 shall automatically reconnect and resume data transmission without restarting the system.

6.3 Security Requirements

NFR-SEC-001 Secure Communication

Communication between the ESP32 and the cloud server should use secure communication protocols such as HTTPS or MQTT over TLS whenever supported.

NFR-SEC-002 User Authentication

Only authorized researchers shall be allowed to access stored experimental data through the dashboard.

NFR-SEC-003 Data Protection

The system shall prevent unauthorized modification of stored experimental records.

6.4 Privacy Requirements

NFR-PRI-001 Audio Privacy

The breathing sensor shall process only breathing sound intensity.

The system shall **not record, store, or transmit speech or voice recordings.**

NFR-PRI-002 Participant Privacy

Experimental data shall be associated with anonymized participant identifiers instead of personally identifiable information.

NFR-PRI-003 Research Usage

Collected data shall be used solely for research, academic analysis, and evaluation of the proposed organic pillow.

6.5 Usability Requirements

NFR-USA-001 Simple Operation

The prototype shall require minimal user interaction during operation.

NFR-USA-002 Dashboard Visualization

Researchers shall be able to visualize recorded sensor data through graphs and comparative reports without requiring specialized software knowledge.

NFR-USA-003 Non-Intrusive Monitoring

The monitoring process shall not significantly interfere with the user's normal sleeping posture or comfort.

6.6 Maintainability Requirements

NFR-MNT-001 Modular Design

Sensor drivers, communication modules, and dashboard components shall be implemented as independent software modules.

NFR-MNT-002 Sensor Replacement

The software architecture shall allow replacement of individual sensors without requiring complete redesign of the application.

NFR-MNT-003 Software Updates

Firmware updates shall be performed independently without modifying the cloud database structure.

6.7 Scalability Requirements

NFR-SCL-001 Sensor Expansion

The system architecture shall permit integration of additional sensors in future versions.

Examples include:

- Humidity Sensor
- Ambient Light Sensor
- Air Quality Sensor
- Additional Pressure Zones

NFR-SCL-002 Multi-User Capability

The database design shall support storage of sleep sessions from multiple participants.

6.8 Portability Requirements

The software shall remain compatible with commonly available ESP32 development boards and standard Wi-Fi networks.

The dashboard shall be accessible from desktop and mobile web browsers.

7. Verification and Validation

Verification and Validation (V&V) activities ensure that the implemented system satisfies the functional and non-functional requirements defined in this SRS.

7.1 Verification Strategy

Verification shall include:

- Hardware inspection
- Firmware testing
- Communication testing
- Dashboard validation
- Experimental evaluation

Each subsystem shall be tested independently before complete system integration.

7.2 Functional Testing

VT-01 Heart Rate Sensor Test

Objective

Verify that the wearable heart rate sensor correctly transmits heart rate values to the ESP32.

Procedure

- Connect the wearable sensor.
- Record heart rate for five minutes.
- Compare displayed values with the wearable device.

Expected Result

Heart rate values are continuously received and displayed without interruption.

VT-02 Breathing Sensor Test

Objective

Verify breathing intensity measurement.

Procedure

- Simulate normal breathing near the microphone.
- Observe changes in measured decibel values.

Expected Result

Breathing waveform responds to inhalation and exhalation.

No speech recordings are stored.

VT-03 Pressure Sensor Test

Objective

Verify Velostat pressure sensing.

Procedure

- Apply pressure at multiple pillow locations.
- Observe pressure distribution.

Expected Result

Pressure values change according to applied force.

Pressure map correctly reflects head position.

VT-04 Head Movement Detection

Objective

Verify movement detection.

Procedure

- Move the head repeatedly across the pillow.
- Compare movement events with observed movements.

Expected Result

Each major movement is detected and logged.

VT-05 Temperature Sensor Test

Objective

Verify temperature measurement.

Procedure

- Warm the pillow surface using normal body contact.
- Observe temperature variation.

Expected Result

Temperature readings increase and gradually decrease after pressure is removed.

VT-06 Cloud Communication Test

Objective

Verify wireless communication.

Procedure

- Connect ESP32 to Wi-Fi.
- Upload sensor data.

Expected Result

Cloud database receives complete sensor records.

VT-07 Dashboard Test

Objective

Verify visualization.

Procedure

- Retrieve stored data.
- Generate graphs.

Expected Result

Dashboard displays all sensor readings correctly.

7.3 System Integration Testing

The complete prototype shall be tested during an overnight sleep session.

The following observations shall be verified:

- Continuous heart rate acquisition
- Continuous breathing monitoring
- Pressure distribution recording
- Head movement detection
- Temperature monitoring
- Wireless communication
- Cloud storage
- Dashboard visualization

The experiment shall be repeated for:

- Conventional pillow
- Proposed organic pillow

The collected datasets shall then be compared.

7.4 Comparative Evaluation

The effectiveness of the proposed organic pillow shall be evaluated by comparing measurements obtained before and after its use.

The following parameters shall be analyzed:

Parameter	Evaluation Method
Heart Rate Trend	Graphical comparison
Breathing Pattern	Waveform comparison
Pressure Distribution	Pressure map analysis
Head Movement Frequency	Number of movement events
Temperature Variation	Temperature profile
User Feedback	Questionnaire responses

7.5 Acceptance Criteria

The proposed prototype shall be considered successful if:

- All sensors operate continuously throughout a sleep session.
- Sensor readings are correctly stored in the cloud database.
- The dashboard displays graphical representations of all recorded parameters.
- Head movement events are successfully detected using the Velostat pressure matrix.
- Breathing intensity is monitored without storing voice recordings.
- Temperature variations are successfully recorded.
- Heart rate measurements are continuously received from the wearable sensor.
- Comparative reports between conventional and organic pillows can be generated.
- Researchers can evaluate differences in movement, pressure distribution, thermal behavior, and user comfort using the recorded data.

7.6 Validation Summary

Requirement Category	Verification Method	Acceptance Standard
Heart Rate Monitoring	Functional Test	Continuous BPM data displayed and stored
Breathing Monitoring	Experimental Test	Breathing waveform detected; no voice recorded
Pressure Monitoring	Load Test	Pressure changes accurately reflected in sensor output
Head Movement Detection	Experimental Observation	Movement events detected and logged
Temperature Monitoring	Thermal Test	Temperature variations recorded correctly
Wi-Fi Communication	Communication Test	Sensor data uploaded successfully to cloud
Dashboard	User Interface Test	Graphs generated without errors
Database	Storage Test	Data stored and retrieved successfully
Comparative Analysis	Experimental Validation	Before-and-after reports generated correctly

APPENDIX A – Acronyms and Abbreviations

Abbreviation	Full Form
ADC	Analog-to-Digital Converter
BLE	Bluetooth Low Energy
BPM	Beats Per Minute
CSV	Comma-Separated Values
ESP32	Espressif 32-bit Wi-Fi and Bluetooth Microcontroller
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
PCB	Printed Circuit Board
REST	Representational State Transfer
RTC	Real-Time Clock
SRS	Software Requirements Specification
TLS	Transport Layer Security
UI	User Interface
Wi-Fi	Wireless Fidelity

APPENDIX B – Glossary

Organic Pillow

A pillow developed using environmentally friendly natural materials such as rice grass, Jatamansi, and processed cow-dung-derived organic material for improved comfort and sustainability.

Sleep Session

A continuous period during which sensor data are collected while a participant sleeps.

Pressure Distribution

The variation of pressure exerted by the user's head on different regions of the Velostat pressure sensing matrix.

Head Movement Event

A significant change in the pressure distribution pattern indicating movement of the user's head during sleep.

Breathing Intensity

The sound pressure level (in decibels) generated by breathing, measured by the microphone without storing audio recordings.

Thermal Behaviour

The heating and cooling characteristics of the pillow surface during and after contact with the user's head.

Comparative Analysis

The process of comparing sensor observations obtained before and after using the proposed organic pillow to evaluate changes in sleep-related parameters.

APPENDIX C – Hardware Specifications

1. ESP32 Development Board

Parameter	Specification
Controller	ESP32
CPU	Dual-Core Xtensa LX6
Clock Speed	Up to 240 MHz
RAM	520 KB SRAM
Flash Memory	4 MB (typical)
Connectivity	Wi-Fi, Bluetooth, BLE

Operating Voltage	3.3 V

2. Wearable Heart Rate Sensor

Parameter	Specification
Type	Wrist-Worn Heart Rate Sensor
Measurement	Heart Rate (BPM)
Interface	Bluetooth Low Energy (BLE)
Usage	Continuous heart rate monitoring

3. Microphone Module

Parameter	Specification
Type	Electret / MEMS Microphone Module
Measurement	Sound Intensity
Output	Analog Signal
Purpose	Breathing pattern monitoring
Privacy	No voice recording or storage

4. Velostat Pressure Matrix

Parameter	Specification
Material	Velostat Conductive Film
Measurement	Pressure Distribution
Output	Variable Resistance
Interface	Analog ADC
Purpose	Head position and movement detection

5. Temperature Sensor

Parameter	Specification
Type	Surface Temperature Sensor
Measurement	Pillow Surface Temperature
Output	Digital/Analog (sensor-dependent)

Purpose	Thermal behaviour monitoring

APPENDIX D – Software Specifications

Component	Technology
Firmware	Arduino IDE (ESP32 Framework)
Programming Language	C++
Database	Firebase Realtime Database / Firestore (depending on implementation)
Dashboard	HTML, CSS, JavaScript
Charts	Chart.js or equivalent
Communication	HTTP or MQTT
Operating System	Windows 10/11

APPENDIX E – System Assumptions

The proposed prototype is developed under the following assumptions:

- The participant wears the heart rate sensor correctly throughout the sleep session.
- The microphone is positioned to capture breathing sounds while minimizing environmental noise.
- The Velostat pressure matrix is properly embedded beneath the pillow surface.
- The temperature sensor maintains contact with the pillow surface throughout the experiment.
- Wi-Fi connectivity is available for cloud synchronization.
- The participant follows the prescribed experimental procedure during both baseline and experimental sessions.
- The collected data are intended for comparative research and not for medical diagnosis.

APPENDIX F – System Constraints

The proposed prototype has the following limitations:

- The system does not diagnose sleep disorders.
- Sleep stages are not estimated.
- Snoring classification is not performed.
- Voice conversations are not recorded or stored.
- The prototype is intended only for comparative research.

- Sensor accuracy depends on proper placement.
- Environmental noise may influence breathing intensity measurements.
- Temperature measurements may vary with room conditions.
- The wearable heart rate sensor is an external device and is not embedded within the pillow.

APPENDIX G – Data Parameters

Parameter	Unit	Source
Heart Rate	BPM	Wearable Sensor
Breathing Intensity	dB	Microphone
Pressure Value	ADC Units	Velostat Matrix
Head Movement Count	Count	Pressure Analysis
Pillow Temperature	°C	Temperature Sensor
Timestamp	Date & Time	ESP32
Session ID	Integer/String	Database
User ID	String	Database

Estimated Cost

Component	Maximum Cost (₹)
ESP32 Development Board	700

MAX30102 Heart Rate Sensor	600
Microphone Module	500
Velostat Sheet	1000
Copper Tape	400
Temperature Sensor (DS18B20/LM35)	600
Jumper Wires & Breadboard	600
Miscellaneous Components	1500
Total Maximum Estimated Cost	**₹6000**

APPENDIX H – Experimental Procedure

The experimental evaluation shall be conducted in two phases.

Phase 1 – Baseline Study

The participant sleeps using a conventional pillow for multiple nights. Sensor readings and questionnaire responses are collected after each session.

Phase 2 – Organic Pillow Study

The participant sleeps using the proposed organic pillow under similar environmental conditions. Identical sensor observations are recorded.

Phase 3 – Comparative Analysis

The collected datasets are compared using graphical visualization and statistical measures to evaluate differences in:

- Heart rate trends
- Breathing intensity
- Pressure distribution
- Head movement frequency
- Pillow surface temperature
- User comfort feedback

APPENDIX I – Dashboard Outputs

The dashboard shall generate the following visualizations:

- Heart Rate Trend Graph
- Breathing Intensity Graph
- Pressure Distribution Heat Map
- Head Movement Timeline
- Temperature Trend Graph
- Session Summary Report
- Comparative Before-and-After Analysis
- Exportable CSV Reports

APPENDIX J – References

1. IEEE Std 29148™-2018, *Systems and Software Engineering—Life Cycle Processes—Requirements Engineering*.
2. IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications* (historical reference).
3. Espressif Systems. *ESP32 Series Datasheet*.
4. Firebase Documentation. *Realtime Database and Cloud Firestore*.
5. Arduino Documentation. *ESP32 Development Framework*.
6. Chart.js Documentation. *Interactive JavaScript Charts*.
7. Relevant peer-reviewed journal articles on smart pillows, IoT-based sleep monitoring, wearable heart-rate sensing, and pressure-based sleep analysis used in the literature review.

Time duration

Task	Duration	Timeline
Literature Survey & Requirement Analysis	1 Week	Week 1
Hardware Procurement & Organic Pillow Preparation	1 Week	Week 2
Hardware Development & Sensor Integration	2 Weeks	Weeks 3–4
System Testing & Calibration	1 Week	Week 5
Baseline Data Collection (Conventional Pillow)	**15 Days**	Weeks 6–7
Experimental Data Collection (Organic Pillow)	**15 Days**	Weeks 8–9
Data Analysis & Comparative Evaluation	1 Week	Week 10
Documentation & Final Report	1 Week	Week 11= 3 Months